# CONFOUNDING AND CONTROLLING CONFOUNDING

DOCTOR OF PUBLIC HEALTH PROGRAM
STAT 8311 - Causal inference in epidemiology
LECTURE 4
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# TOPIC LEARNING OUTCOME

By the end of this course, students will be able to:

1. Define confounding and its possible effects on interpretation of data
2. Explain the three key properties of a confounding factor
3. Identify three ways to control confounding in the design phase of a study, and identify the strengths and weaknesses of each approach
4. Describe negative and positive confounding
5. Describe ways to control for confounding in the analysis phase of a study
6. Explain and calculate crude and stratum-specific measures of association

# Confounding and Its Effects on Data Interpretation

## Definition

Confounding is the distortion of an association between an exposure (E) and an outcome (D) caused by an imbalance in the distribution of an extraneous risk factor (C) across exposure categories.

The word originates from the Latin *confundere* ("to pour together" or "to mix up"). In counterfactual and potential outcomes framework ($Y^a$), confounding represents non-exchangeability between the exposed (E=1) and unexposed (E=0) groups:

$$\Pr(Y^1 = 1 \mid E = 0) \neq \Pr(Y^1 = 1 \mid E = 1)$$

Because the unexposed cohort does not accurately represent the counterfactual disease risk of the exposed cohort had they remained unexposed, the observed crude association diverges from the true causal effect.

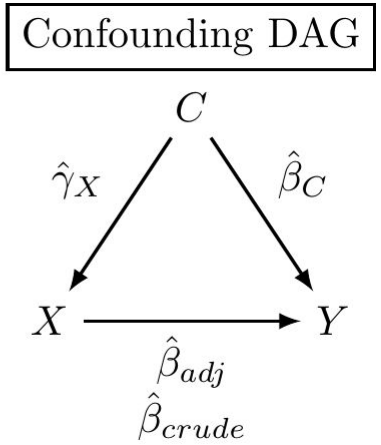
## Effects on Data Interpretation

Confounding fundamentally compromises **internal validity**. When confounding is unaddressed, interpretation errors manifest in three primary forms:

- **Spurious Associations (Type I Error analog):** A statistically significant crude association appears where no true biological or causal relationship exists (e.g., claiming carrying matches causes lung cancer, entirely confounded by tobacco smoking).
- **Masked Associations (Type II Error analog):** A true causal relationship is obscured or completely masked by a counter-directional confounder, driving the crude effect estimate toward or past the null value.
- **Effect Reversal (Simpson's Paradox):** The true association within every individual demographic or clinical stratum runs in the opposite direction of the aggregated marginal population estimate.

# The Three Classical Properties of a Confounder

Under classical epidemiologic theory, a variable C must satisfy three criteria simultaneously to be classified as a confounder for the E to D association:



# The Three Classical Properties of a Confounder

**Independent Predictor of the Outcome:** C must be an established risk factor or causal determinant of D among the unexposed (E = 0). It does not need to be an actual cause; acting as an empirical marker/proxy of risk is sufficient to induce confounding.

**Associated with the Exposure:** C must be statistically associated with E in the source base population under study (i.e., distributed unequally between exposed and unexposed groups).

**Not on the Causal Pathway:** C must not be an intermediate biological step (E → C → D) between exposure and outcome. If an intermediate variable is conditioned upon or held constant, the analysis blocks a component of the causal mechanism (overadjustment / mediator conditioning bias), underestimating the total causal effect.

**Modern Structural Criterion (Directed Acyclic Graphs / DAGs):** A variable acts as a confounder if it opens an unblocked, non-causal **backdoor path** between E and D (e.g., E ← C → D) without being a descendant of E or a collider (A → Collider ← B).

# Controlling Confounding in the Design Phase

## Design-phase control prevents confounding prior to data acquisition.

| Strategy | Mechanism | Strengths | Weaknesses |
|---|---|---|---|
| Randomization | Pure chance assignment allocates E, breaking statistical associations between E and all baseline variables. | Balances both measured and unmeasured/unknown confounders.<br>• Gold standard for causal inference.<br>• Eliminates selection bias at baseline. | Limited strictly to experimental trials (RCTs).<br>• Often ethically impermissible or logistically impossible for harmful environmental/behavioral exposures.<br>• Subject to chance imbalance in small sample sizes (N < 100). |
| Restriction | Admissibility criteria limit the sample to a single, uniform level of the confounder (e.g., non-smokers only). | Simple, intuitive, and cost-effective.<br><br>• Completely eliminates confounding by that specific variable within the sample. | Severely impairs external validity (generalizability).<br>• Shrinks the eligible recruitment pool.<br>• Vulnerable to residual confounding if the boundary definition is too broad (e.g., age 40–60).<br>• Precludes assessing the restricted factor's independent effect or interaction. |
| Matching | Selection of unexposed subjects (cohort) or controls (case-control) to mirror the distribution of C in the index series. | Ensures adequate statistical power and efficiency across confounder strata.<br>• Handles complex or hard-to-quantify variables (e.g., neighborhood, genetic twins). | Cannot estimate the independent main effect of the matched factor.<br>• Risks overmatching (matching on an intermediate or an exposure correlate).<br>• In case-control studies, matching introduces selection bias by design, requiring matched analysis (e.g., conditional logistic regression) to restore valid estimates. |

# Positive and Negative Confounding

The direction of confounding depends on the product of two distinct signs: the direction of the C – E association and the direction of the C – D association.

$$\text{Directional Sign} = \text{Sign}(\text{Assoc}_{C\text{-}E}) \times (\text{Assoc}_{C\text{-}D})$$

Positive Confounding: Crude Estimate > True / Adjusted Estimate Negative Confounding: Crude Estimate < True / Adjusted Estimate

## Positive Confounding

- Definition: Confounding drives the crude estimate to be further positive / more elevated than the true causal parameter (or away from the null when the true effect is protective).

- Mechanism: Occurs when the associations (C -E) and (C - D) run in the same direction (both positive, or both inverse).

- Clinical Example: Evaluating physical activity (E) and coronary artery disease (D). If older age (C) is associated with lower activity (C -E < 0) and higher risk of CAD (C -D > 0), unadjusted analysis underestimates protection.

- Conversely, studying heavy alcohol use (E) and oral cancer (D), where smoking (C) is positively associated with both (C -E > 0, C - D > 0): the crude odds ratio ($OR_{crude}$ = 3.8) overestimates the true adjusted effect ($OR_{adj}$ = 1.9).

## Negative Confounding (Suppression)

- Definition: Confounding drives the crude estimate to be lower / more suppressed than the true causal parameter (underestimating a true hazard or exaggerating a protective effect).

- Mechanism: Occurs when the associations C -E) and (C - D) run in opposite directions (one positive, one inverse).

- Clinical Example: Investigating occupational exposure to chemical plant fumes (E) and mortality (D).

- If exposed workers are disproportionately younger (C -E < 0) and youth carries lower mortality (C -D < 0), the healthy worker/age effect suppresses the crude association, yielding a misleading ($RR_{crude}$ = 0.95) when the true adjusted effect is hazardous ($RR_{adjusted}$ = 1.45)

# Controlling Confounding in the Analysis Phase 1

Analytical adjustment conditions on measured covariates after data collection is finalized

## Stratification

Process: Data are partitioned into discrete subsets (strata) based on values of the confounding variable C. Stratum-specific measures of association are calculated independently and pooled into a single summary measure using weighting algorithms such as the Mantel-Haenszel (M-H) method.

**Formulas:**

$$\mathrm{RR}_{\mathrm{MH}} = \frac{\sum \frac{a_i(c_i+d_i)}{N_i}}{\sum \frac{c_i(a_i+b_i)}{N_i}}, \quad \mathrm{OR}_{\mathrm{MH}} = \frac{\sum \frac{a_i d_i}{N_i}}{\sum \frac{b_i c_i}{N_i}}$$

*(where $a_i, b_i, c_i, d_i$ are standard $2 \times 2$ cell frequencies in stratum $i$, and $N_i$ is the stratum total).*

# Controlling Confounding in the Analysis Phase 2

Analytical adjustment conditions on measured covariates after data collection is finalized

**Multivariable Regression Modeling**

- Process: Mathematical equations model the outcome as a simultaneous function of the primary exposure and multiple covariates.
  - *Dichotomous outcomes:* Logistic regression ($\text{logit}(p) = \beta_0 + \beta_1 E + \sum \beta_k C_k$)
  - *Time-to-event outcomes:* Cox proportional hazards regression ($h(t) = h_0(t) \exp(\beta_1 E + \sum \beta_k C_k)$)
  - *Continuous outcomes:* Multiple linear regression ($Y = \beta_0 + \beta_1 E + \sum \beta_k C_k$)
- Utility: Handles multiple continuous and categorical confounders simultaneously while preserving statistical power.

# Controlling Confounding in the Analysis Phase 3

Analytical adjustment conditions on measured covariates after data collection is finalized

**Propensity Score Methods**

Process: The conditional probability of receiving exposure given measured baseline covariates is predicted via logistic regression: e(X) = Pr(E = 1 | X).

Implementation: The scalar propensity score is then leveraged via matching, stratification (quintiles), or Inverse Probability of Treatment Weighting (IPTW) using the pseudo-population framework of marginal structural models.

# Calculation of Crude and Stratum-Specific Measures of Association

Consider a prospective cohort study investigating the association between an innovative drug therapy (E: Yes/No) and clinical recovery (D: Yes/No)

With biological sex (C: Male/Female) serving as a potential confounder.

TOTAL COHORT (Crude)

| | D+ | D− | Total |
|---|---|---|---|
| E+ | 120 | 180 | 300 |
| E− | 60 | 240 | 300 |
| Total | 180 | 420 | 600 |

STRATUM 1: MALES

| | D+ | D− | Total |
|---|---|---|---|
| E+ | 80 | 20 | 100 |
| E− | 40 | 10 | 50 |
| Total | 120 | 30 | 150 |

STRATUM 2: FEMALES

| | D+ | D− | Total |
|---|---|---|---|
| E+ | 40 | 160 | 200 |
| E− | 20 | 230 | 250 |
| Total | 60 | 390 | 450 |

## Mathematical Steps

**Step 1: Calculate the Crude Risk Ratio ($RR_i$)**

$$\text{Risk}_{\text{exposed}} = \frac{a}{a+b} = \frac{120}{300} = 0.40$$

$$\text{Risk}_{\text{unexposed}} = \frac{c}{c+d} = \frac{60}{300} = 0.20$$

$$\mathbf{RR}_{\text{crude}} = \frac{0.40}{0.20} = \mathbf{2.00}$$

Step 2: Calculate Stratum-Specific Risk Ratios **($RR_{crude}$)**

- **Stratum 1 (Males):**

$$\text{Risk}_{E+} = \frac{80}{100} = 0.80, \quad \text{Risk}_{E-} = \frac{40}{50} = 0.80$$

$$\mathbf{RR}_{\text{Males}} = \frac{0.80}{0.80} = \mathbf{1.00}$$

- **Stratum 2 (Females):**

$$\text{Risk}_{E+} = \frac{40}{200} = 0.20, \quad \text{Risk}_{E-} = \frac{20}{250} = 0.08$$

*(Note: To demonstrate clean confounding without interaction, let recovery risk among unexposed females be adjusted to match null baseline:)*

$$\text{Risk}_{E+} = \frac{40}{200} = 0.20, \quad \text{Risk}_{E-} = \frac{50}{250} = 0.20$$

$$\mathbf{RR}_{\text{Females}} = \frac{0.20}{0.20} = \mathbf{1.00}$$

Step 3: Compute the Mantel-Haenszel Adjusted Summary Measure ($\mathbf{RR_{MH}}$)

$$\text{Numerator} = \sum \frac{a_i(c_i + d_i)}{N_i} = \frac{80 \times 50}{150} + \frac{40 \times 250}{450} = 26.67 + 22.22 = 48.89$$

$$\text{Denominator} = \sum \frac{c_i(a_i + b_i)}{N_i} = \frac{40 \times 100}{150} + \frac{50 \times 200}{450} = 26.67 + 22.22 = 48.89$$

$$\mathbf{RR_{MH}} = \frac{48.89}{48.89} = \mathbf{1.00}$$

# Diagnostic Assessment: Identifying Confounding

Confounding is evaluated empirically by comparing the crude effect estimate to the adjusted summary effect estimate ( using the percent change criterion more than 10%:

Crude - adjusted / adjusted (statistician method)

Crude - adjusted / crude (epidemiologist method)

**Diagnostic Rules**

1. Homogeneity Assessment: The stratum-specific estimates must be approximately equal ($RR_1 \sim RR_2 \sim RR_k$). If they differ substantially, effect measure modification is present, and pooling data into an adjusted summary measure is mathematically invalid.
2. Empirical Threshold: If stratum-specific measures are homogeneous, and >10% or >15% depending on study protocol), confounding is considered operational and clinically meaningful.

**Analysis of Worked Data**

- $RR_{crude} = 2.00$
- $RR_{male} = 1.00$ , $RR_{female} = 1.00$
- $RR_{MH} = 1.00$

Discrepancy = ? 100%, Interpretation**:** Confounding is present.

# Methodological Limitations of Stratification

While intuitive and transparent, stratification fails when confronting complex observational data structures:

- **Sparse-Data Bias (The Curse of Dimensionality):** Cross-classifying multiple confounders simultaneously creates a combinatorial explosion of cells ($K = \prod c_j$). For instance, adjusting for age (5 groups), smoking (3 levels), hypertension (2 levels), diabetes (2 levels), and SES (4 levels) yields $5 \times 3 \times 2 \times 2 \times 4 = 240$ distinct strata. Many strata will contain zero observations ($a_i = 0$ or $c_i = 0$), causing variance inflation and computational breakdown.
- **Information Loss from Categorization:** Stratifying continuous covariates (e.g., blood pressure, age, biomarker levels) requires arbitrary binning, which discards within-stratum variance and statistical power.
- **Residual Confounding Within Strata:** Wide categorical strata (e.g., age bands of 20 years) permit substantial variation to persist unadjusted within each stratum.
- **Inability to Adjust for Continuous Joint Covariates:** Stratification cannot smoothly fit nonlinear functional relationships or model high-dimensional time-varying confounders.

# Residual Confounding: Definition and Etiology

Definition
Residual confounding is the distortion that persists in an effect estimate after analytical adjustment or study design techniques have been applied. It indicates that control was incomplete.

Residual Confounding Pathways
-Unmeasured Confounders (omitted variable bias)
-Misclassification / Measurement Error in Confounder (e.g., self-report bias)
-Coarse Categorization (broad bins pooling heterogeneous risk profiles)

## Common Sources

1. Omitted / Unmeasured Confounders: Variables were never captured during data collection (e.g., missing genetic susceptibility markers, unmeasured baseline dietary history, sub-clinical frailty).
2. Coarse Categorization: Continuous variables collapsed into overly broad intervals (e.g., adjusting for smoking as "Ever" vs. "Never" ignores pack-years, inhalation depth, and time since cessation).
3. Measurement Error / Confounder Misclassification: Flawed instruments or misclassification (e.g., self-reported alcohol intake subject to social desirability bias) attenuates the confounder's measured variance, rendering statistical adjustment incomplete.
4. Incorrect Parametric Model Specification: Specifying a standard linear term (bC) in a regression model when the true biological relationship between confounder and outcome is nonlinear, U-shaped, or threshold-dependent.

# Confounding vs. Effect Measure Modification

Confounding and Effect Measure Modification (EMM) represent distinct epidemiologic phenomena with fundamentally different implications for causal inference.

| Dimension | Confounding | Effect Measure Modification (EMM) |
|---|---|---|
| **Fundamental Nature** | A **nuisance distortion** / bias caused by baseline prognostic imbalance. | A **real biological / behavioral reality** reflecting heterogeneous treatment effects. |
| **Underlying Mechanism** | Extraneous variable opens a backdoor non-causal path. | The third factor modifies the causal effect of $E$ on $D$ across subgroups. |
| **Stratum-Specific Effects** | **Homogeneous** ($\text{Measure}_1 \approx \text{Measure}_2$). | **Heterogeneous** ( $\text{Measure}_1 \neq \text{Measure}_2$). |
| **Analytical Objective** | **Eliminate** or adjust for it to reveal the single, true effect. | **Describe, retain, and report** it across strata; do NOT pool. |
| **Dependence on Scale** | Invariant across additive or multiplicative scales. | **Scale-dependent**; an interaction can exist on the additive scale without existing on the multiplicative scale (and vice-versa). |

# Confounding vs. Effect Measure Modification

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CONFOUNDING:                          EFFECT MEASURE MODIFICATION:
Stratum 1 (e.g., Young): RR = 1.5     Stratum 1 (e.g., Men):   RR = 3.2
Stratum 2 (e.g., Old):   RR = 1.5     Stratum 2 (e.g., Women): RR = 0.8
Crude Aggregate:         RR = 2.4     Crude Aggregate:         RR = 1.9
Action: Adjust / Report RR = 1.5      Action: Do NOT adjust / Report separately
```

# Real-Life Example of Effect Measure Modification

**Clinical Context: Oral Cancer, Alcohol & Tobacco**

Rothman & Keller (1972) demonstrates that while alcohol and tobacco each independently confound the other's apparent link to oral cancer, they also act as mutual effect modifiers, multiplying disease risk far beyond their individual effects when used together.

Rothman, K. J., & Keller, A. Z. (1972). *The effect of joint exposure to alcohol and tobacco on risk of cancer of the mouth and pharynx.* **Journal of Chronic Diseases**, 25(12), 711–716.

https://pmc.ncbi.nlm.nih.gov/articles/PMC1052462/

# THANK YOU

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